Physical Chemistry CUME, January 26, 2018

Quantum Chemistry

1. An electron moving in the y direction is in an eigenstate of $\hat{S}_z$ operator with eigenvalue $\frac{\hbar}{2}$ before it passes through a Stern-Gerlach analyzer that is oriented in the $+x$ direction. What is the probability of finding the outgoing electron in the spin up state (along x)? [10 points]

2. In Problem 1, suppose that the Stern-Gerlach analyzer is oriented along $\hat{\boldsymbol{n}}$, where $\hat{\boldsymbol{n}}$ is $\left(-\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right)$. What is the probability of finding the outgoing electron in the spin up state (along $\hat{\boldsymbol{n}}$)? [30 points]

3. A system consisting of a pair of electrons is in a singlet state ($|0,0\rangle$). Write down the normalized spin wavefunction of this system in terms of $|\uparrow\rangle_1$, $|\downarrow\rangle_1$, $|\uparrow\rangle_2$, and $|\downarrow\rangle_2$. [10 points]

4. In Problem 3, suppose that the two electrons are not locally bound and move in opposite directions along y such that they can pass through separate Stern-Gerlach analyzers at the same time. What are the possible outcomes of the two measurements (and their probabilities), if both Stern-Gerlach analyzers are oriented in the $+z$ direction? [15 points]

5. In Problem 4, discuss the possible outcomes and their probabilities, if the two Stern-Gerlach analyzers are oriented perpendicular to each other along $+x$ and $+z$ directions, respectively? [15 points]

6. In Problem 4, discuss the possible outcomes and their probabilities, if the system is in a triplet state, where the spin wavefunction is described by $\frac{1}{2}\left(|1,-1\rangle + \sqrt{2}|1,0\rangle + |1,1\rangle\right)$. [20 points]